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An integrated clinical-histopathologic prediction model for cardiac allograft rejection: Translating machine learning into clinical risk frameworks

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KEYWORDS:

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Machine learning and
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Digital pathology;
Risk prediction;
Heart transplantation

BACKGROUND: Cardiac allograft rejection (CAR) remains the leading cause of early graft failure after heart transplantation (HT). Current diagnostics, including histologic grading of endomyocardial biopsy (EMB) and blood-based assays, lack accurate predictive power for future CAR risk. We developed a predictive model integrating routine clinical data with EMB-derived quantitative morphologic features to demonstrate the precision medicine potential of mining existing data sources in post-HT care.

METHODS: In a retrospective cohort of 484 HT recipients with 1,188 EMB encounters within 6 months post-transplant, we extracted 370 quantitative pathology features from digitized H&E-stained slides and 268 longitudinal clinical features from routine lab testing. Trained using the XGBoost algorithm, we compared model performance across time (cross-sectional vs longitudinal) and across data domains (clinical vs morphologic). The top predictors of our best-in-class model informed the derivation of a simplified Integrated Rejection Risk Index (IRRI) for patient risk stratification. Model performance was evaluated by AUROC, AUPRC, and time-to-event hazard ratios.

RESULTS: The fully integrated longitudinal model, inclusive of clinical and morphologic domains, achieved superior predictive accuracy (AUROC 0.86, AUPRC 0.74) compared to cross-sectional or single-domain models. IRRI stratified patients into risk categories with distinct future CAR hazards: high-risk patients showed a markedly increased CAR risk (HR = 6.15, 95% CI: 4.17-9.09), while low-risk patients had significantly reduced risk (HR = 0.52, 95% CI: 0.33-0.84).

CONCLUSIONS: By integrating longitudinal clinical and biopsy morphologic features, IRRI provides a scalable, interpretable tool for proactive CAR risk assessment. This precision-based approach offers a promising pathway toward risk-adaptive surveillance and immunosuppressive strategies.

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Non-standard Abbreviations and Acronyms: CAR, cardiac allograft rejection; CfDNA, cell-free DNA; GEP, gene expression profiling; IS, immunosuppressant

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Background

Heart transplantation (HT) is the treatment of choice for end-stage heart failure, with more than 6,000 transplants performed worldwide in 2024.¹ While HT is the most effective and durable therapy for patients with end-stage HF,²

the recipient immune response is a persistent threat to allograft health, necessitating vigilant surveillance and life-long immunosuppression (IS) to mitigate the risk of cardiac allograft rejection (CAR).^{3–5}

The care HT patients receive is highly regimented, with the schedule for CAR surveillance testing and IS drug de-escalation determined by local institutional protocols. Historically, endomyocardial biopsy (EMB) with histologic grading had been the sole diagnostic standard for CAR and has been recommended by HT guidelines since 1990 despite concerns about inter-grader reliability and accuracy.^{6–8} More recently, ‘liquid biopsy’ methods using plasma-based cell-free DNA (cfDNA) and gene expression profiling have been developed as less-invasive CAR screening alternatives for select patients, with positive results triggering confirmatory EMB.^{9–13} Regardless of approach, it is typical for an HT recipient to undergo 12 or more scheduled surveillance CAR tests and 12 or more scheduled changes in IS regimens during their first year post-transplant alone.¹⁴

Due to “one-size-fits-all” protocols, patients at low risk for allograft complications undergo more diagnostic testing and receive more aggressive IS than necessary, compounding immunosuppressive health risks while wasting scarce healthcare resources.¹⁵ Despite clear benefits to patients, cfDNA testing has a high false-positive rate (>90%) and thus still necessitates frequent EMBs.^{12,16} And, though generally safe, EMBs still introduce risk in addition to adding additional costs.^{17–19} Rigid, reactive protocols pose an even greater threat to patients at high risk for future allograft complications who may experience inadequate surveillance and premature IS weaning. Furthermore, the extra expense and provider time required to manage these CAR-related complications represent a form of potentially preventable resource waste in post-HT care.^{20,21} Ultimately, there are clear inefficiencies in the post-HT setting which compromise both the quality of care patients receive and the allocation of resources for transplant programs and health systems.

Fundamentally, these inefficiencies reflect a lack of reliable, proactive, individualized CAR risk assessment tools. Neither the established EMB histologic grading criteria nor the more recently introduced cfDNA approaches are designed to predict rejection events weeks or months in the future.¹⁶ While there have been limited studies in HT analyzing trends between certain clinical biomarkers and acute²² or chronic CAR,²³ these methods have not been adopted clinically. More recently, efforts to utilize computational analysis of digitized EMB slides have yielded some promising results for improving the diagnostic value of EMB,^{24–27} but have not yet been utilized to predict future CAR events. In this study, we generate an integrated clinical and morphologic model for prediction of future acute and subacute CAR after HT, providing a decision-support framework for precise, patient-centered care in HT.

Methods

Study cohort and design

We conducted a retrospective study of adult HT recipients at the University of Pennsylvania using a cohort derived

from 4 previously published datasets of digitized EMB slides.^{24–27} The analytic cohort was restricted to patients with at least 2 digitized H&E-stained EMB obtained within the first 6 months post-transplant and with sufficient linked electronic health record (EHR) data to enable longitudinal feature extraction. To focus on outpatient surveillance and minimize perioperative confounding, EMBs and clinical data from the first 20 days post-transplant were excluded. Patients were additionally required to have ≥90% completeness of routinely collected laboratory and clinical variables; residual missingness was addressed using multiple imputation by chained equations (MICE—see [Supplemental Methods](#)).²⁸ The final cohort comprised 484 patients, contributing 1,188 EMB-associated outpatient encounters. Not all clinically performed biopsies were digitized or available for analysis. Digitized EMBs were preferentially available at key early post-transplant time-points (typically ~1 and ~3–4 months), reflecting the structure of source digital slide cohorts rather than complete biopsy capture.

The objective of this study was to determine whether future CAR risk can be predicted using routinely acquired post-transplant data, and whether longitudinal and multi-modal strategies outperform cross-sectional (eg. time of visit) or single-domain approaches. We therefore adopted a 2-stage experimental design. First, we assessed feasibility by training high-dimensional machine-learning models to predict short-term CAR occurring > 7 days and < 6 weeks after an outpatient encounter. This time horizon was selected based on both biological and clinical considerations: prior work from our group using multiplex immunofluorescence demonstrated divergence in biopsy immune profiles within approximately 1–2 months prior to clinically evident rejection, suggesting that this interval may represent a biologically meaningful window for early risk detection, while also aligning with the cadence of outpatient follow-up during the first 6 months post-transplant. Second, we evaluated translational potential by distilling the most informative features into a constrained, interpretable risk index designed to stratify medium-term CAR risk over a 90-day horizon, a timeframe selected to better reflect clinically actionable surveillance and IS decisions.

Outcomes

For the “feasibility” modeling task, encounters were labeled based on whether a CAR event occurred between > 7 days and ≤42 days following a CAR-free patient encounter. CAR was defined as ISHLT acute cellular rejection (grade > 1R), pathologic antibody-mediated rejection (pAMR > 0), or “clinically evident” rejection (as defined previously²⁷). For translational risk index development, patients were classified according to whether any CAR event occurred within 90 days of an encounter. Patients experiencing such an event were labeled Future Rejection (FR, $n = 134$ patients, $n = 320$ encounters), while those without events were labeled No Rejection (NR, $n = 350$ patients, $n = 868$ encounters). Among FR patients, EMBs were collected a mean of 39.75 (SD ± 29.9)

and median of 29.0 days before rejection. The analytical cohort from which the digitized slides were derived was modestly enriched for FR patients (26.9% vs 15.1% overall institutional incidence) to mitigate class imbalance and ensure adequate statistical power for predictive modeling. In real-world post-transplant populations, rejection events are relatively infrequent, and highly imbalanced datasets can adversely affect classifier training and limit the ability to detect meaningful predictive signals. Enrichment of the minority class is a common strategy in supervised learning to address this limitation²⁹ and has been used in prior transplant computational pathology publications.^{25,26} As a result, the analytical cohort reflects a higher event prevalence than would be expected in routine clinical practice.

Feature extraction and dataset construction

Each outpatient encounter served as a unit of analysis, with clinical, demographic, and morphologic features extracted as “cross-sectional” variables for each encounter. Longitudinal variables were constructed using structured time bins and included change-from-baseline, change-from-prior encounter, and rate-of-change features, with time-varying inputs represented as engineered summary features rather than explicit time-series sequences. This approach was selected to preserve interpretability and facilitate translation into a simplified clinical risk framework, rather than relying on more complex sequential modeling approaches that generate less interpretable latent features. To account for repeated encounters within patients, all model training and evaluation were performed using patient-level data splits, ensuring that encounters from a given patient were not shared between the training and test sets.

Quantitative morphology data were extracted from digitized H&E EMBs using previously validated automated digital pathology pipelines which assess immune cell burden, stromal remodeling, and tissue architecture metrics.^{24–27} Figure 1 provides visual examples of quantitative morphology analysis (refer to the Supplement for detailed methods). Clinical features included laboratory values, medication exposures (medications started, continued, or discontinued), and demographic variables and were structured as cross-sectional, baseline, and/or longitudinal change-over-time measures (Figure S.1 and [Supplemental Methods](#)). All features were concatenated into a single analytic matrix comprising 1,361 variables per encounter (Table S.2).

Hypotheses and predictive modeling

Predictive modeling was designed to test 3 primary hypotheses: (1) longitudinal features outperform cross-sectional data for predicting future CAR; (2) integration of quantitative morphologic features enhances predictive performance beyond conventional clinical and histologic inputs; and (3) predictive information can be distilled into a small number of interpretable risk factors suitable for clinical use.

To address hypotheses 1 and 2, we trained gradient-boosted decision tree classifiers using XGBoost. Data were split 80:20 into training and test sets by patient identifier to prevent information leakage, with 10-fold cross-validation applied during model training. Model performance was evaluated exclusively on the held-out test set using the area under the receiver operating characteristic curve (AUROC) and the area under the precision–recall curve (AUPRC).

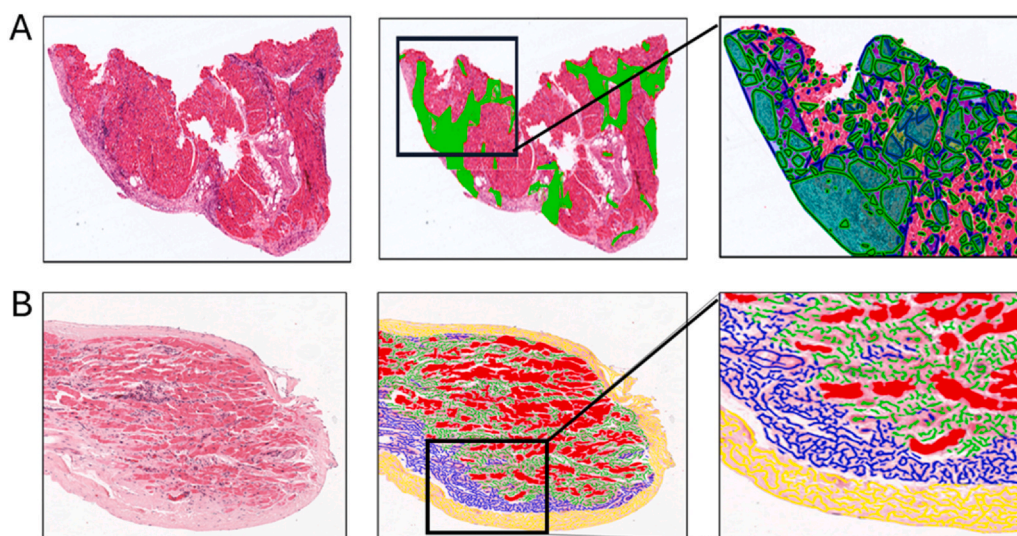


Figure 1 Quantitative morphology feature extraction examples. Digitized H&E whole-slide-images are analyzed by the lymphocyte feature pipeline (A) which identifies and measures lymphocyte colonies and foci and quantifies their relationship to surrounding myocyte structures, and by the stromal feature pipeline (B) which identifies stromal fibers in the digitized H&E slides, separating them into endocardial stroma, interstitial stroma (e.g., closely bounding cardiomyocytes), and stromal “islands” which are macrostromal domains which are less closely associated with cardiomyocytes. The stromal fibers are then measured by area, orientation, and other geometrics and relational properties.

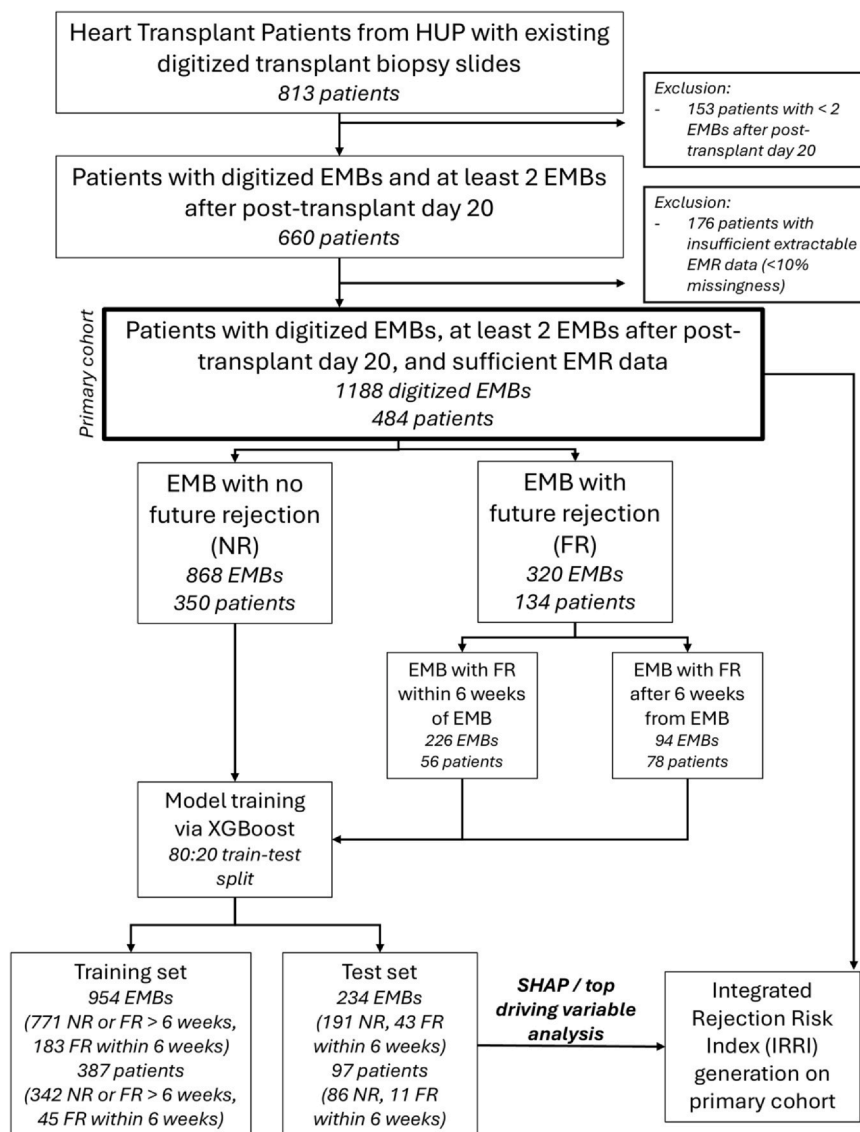


Figure 2 Consort diagram of patient cohort. Inclusion criteria for the primary cohort were patients with associated digitized EMBs and documented laboratory and medication data.

Development of the integrated rejection risk index

To facilitate clinical translation and directly address hypothesis 3, we derived the Integrated Rejection Risk Index (IRRI) by selecting influential and routinely available predictors identified through SHAP value rankings^{30,31} from the XGBoost model. Selected features were grouped into “clinical” and “morphologic” domains, and composite scores were generated to stratify outpatient encounters into low-, intermediate-, and high-risk categories (study workflow summarized in Figure 2).

The association between IRRI risk strata and subsequent CAR was assessed using Kaplan-Meier time-to-event analysis and Cox proportional-hazards modeling. Hazard ratios with 95% confidence intervals were calculated to quantify differential risk across strata, with log-rank testing used to assess separation of rejection-risk trajectories. Cox proportional hazards models were used to assess the association between IRRI risk strata and time to subsequent CAR. Proportional hazards assumptions were evaluated using Schoenfeld residuals, with no significant violations detected.

Results

Cohort characteristics

Recipient and donor demographics and characteristics stratified by rejection status are described in Table 1. Of the examined features, there were no statistically significant differences between the FR and NR cohorts for recipient or donor baseline characteristics.

Predictive modeling

Longitudinal clinical data enhances predictive performance over cross-sectional approaches

Using only routinely available, cross-sectional clinical data ($n = 185$ variables) from study outpatient encounters, the XGBoost model achieved an AUROC of 0.62 and an AUPRC of 0.28 for predicting CAR within 6 weeks of the

Table 1 Descriptive Statistics of Recipients and Donor Baseline Characteristics, Stratified by Whether the Recipient is a “Future Rejector” (FR) or “Never Rejector” (NR)

		Grouped by NR / FR designation				
	Missing	Group	Overall	FR	NR	<i>p</i> -value
<i>n</i>			484	134	350	
RECIPIENT CHARACTERISTICS						
Age at transplant, mean (SD)	0		52.6 (12.3)	53.9 (11.6)	52.1 (12.5)	0.141
Male, <i>n</i> (%)	0		373 (77.1)	97 (72.4)	276 (78.9)	0.163
Race, <i>n</i> (%)	0	White	342 (70.7)	96 (71.6)	246 (70.3)	0.052
		Black	93 (19.2)	18 (13.4)	75 (21.4)	
		Other	49 (10.1)	20 (14.9)	29 (8.3)	
BMI at Transplant, mean (SD)	120		27.4 (5.2)	26.8 (4.6)	27.6 (5.4)	0.155
Redo Transplant, <i>n</i> (%)	0		13 (2.7)	3 (2.2)	10 (2.9)	1.000
CMV status, <i>n</i> (%)	0		221 (45.7)	56 (41.8)	165 (47.1)	0.339
Baseline creatinine, mean (SD)			1.4 (1.1)	1.3 (0.7)	1.4 (1.2)	0.057
Baseline eGFR, mean (SD)	0		67.1 (28.8)	66.0 (31.5)	67.5 (27.8)	0.620
LVAD hx, <i>n</i> (%)	0		34 (7.0)	10 (7.5)	24 (6.9)	0.972
Occlusive CAD hx, <i>n</i> (%)	0		166 (34.3)	43 (32.1)	123 (35.1)	0.599
Ischemic cardiomyopathy hx, <i>n</i> (%)	0		78 (16.1)	25 (18.7)	53 (15.1)	0.422
Hx of Diabetes, <i>n</i> (%)	0		127 (26.2)	43 (32.1)	84 (24.0)	0.090
Hx pulmonary hypertension, <i>n</i> (%)	0		103 (21.3)	33 (24.6)	70 (20.0)	0.323
Hx of Hypertension, <i>n</i> (%)	0		174 (36.0)	51 (38.1)	123 (35.1)	0.622
Hx of Hyperlipidemia, <i>n</i> (%)	0		171 (35.3)	43 (32.1)	128 (36.6)	0.414
Hx of CKD, <i>n</i> (%)	0		92 (19.0)	25 (18.7)	67 (19.1)	1.000
cPRA, <i>n</i> (%)	0	0%	461 (95.2)	125 (93.3)	336 (96.0)	0.354
		0%-19%	9 (1.9)	3 (2.2)	6 (1.7)	
		20%-49%	7 (1.4)	2 (1.5)	5 (1.4)	
		> 50%	7 (1.4)	4 (3.0)	3 (0.9)	
DONOR CHARACTERISTICS						
Age of Donor, mean (SD)	0		37.7 (13.4)	38.4 (13.8)	37.4 (13.3)	0.449
Male Donor, <i>n</i> (%)	0		337 (69.6)	92 (68.7)	245 (70.0)	0.859
Donor LVEF, mean (SD)			59.8 (11.4)	59.9 (11.1)	59.7 (11.6)	0.820
Donor BMI, mean (SD)	0		27.9 (6.1)	28.7 (6.2)	27.7 (6.1)	0.098
Donor eGFR, mean (SD)	2		83.6 (48.1)	80.7 (50.0)	84.6 (47.3)	0.436
Donor ischemic time	0		3.0 (0.9)	3.0 (0.9)	3.1 (0.9)	0.299
Donor hx of cigarette use, <i>n</i> (%)	0		123 (25.4)	33 (24.6)	90 (25.7)	0.897
Donor hx of HTN, <i>n</i> (%)	0		373 (77.1)	100 (74.6)	273 (78.0)	0.503
Donor hx of diabetes, <i>n</i> (%)	0		111 (22.9)	34 (25.4)	77 (22.0)	0.104

*Abbreviations: BMI, body mass index; CAD, coronary artery disease; CKD, chronic kidney disease; CMV, cytomegalovirus; cPRA, calculated panel reactive antibody; eGFR, estimated glomerular filtration rate; HTN, hypertension; hx, history; LVAD, left ventricular assist device; LVEF, left ventricular ejection fraction.

encounter (F1: 0.57; Sensitivity: 0.53; Specificity: 0.70) (Figure 3A–B, Table 2). When longitudinal variables—such as percent change from first and most recent outpatient encounter, and change-over-time metrics—are integrated into the feature space ($n = 268$ variables total), predictive performance improved substantially, with an improved AUROC of 0.77 ($p = 0.007$) and an improved AUPRC of 0.54 ($p = 0.0038$) (F1: 0.70; Sensitivity: 0.47; Specificity: 0.91) (Figure 3A–D, Table 2).

Multi-modality integration further improves prediction

Building on prior evidence of histopathologic utility in HT rejection assessment,^{24–27} we next evaluated whether integrating quantitative EMB-derived morphologic features

with clinical data could further improve prediction of future CAR.

The integrated clinical–morphologic model with longitudinal variables outperformed the clinical-only model with longitudinal variables, achieving AUROC of 0.86 and an AUPRC of 0.74 for predicting CAR within 6 weeks of an encounter (Figure 4A–B, Table 2). Performance gains were significant for AUPRC ($p = 0.029$) and directionally consistent for AUROC ($p = 0.057$) compared to the conventional clinical data model. Across all tested configurations, the integrated model yielded the highest overall performance metrics (F1 = 0.83; Sensitivity = 0.72; Specificity = 0.94, Figure 4C), with parameters for all models summarized in Table S3.

SHAP analysis demonstrated that morphologic features contributed the largest effect sizes, driven primarily by stromal

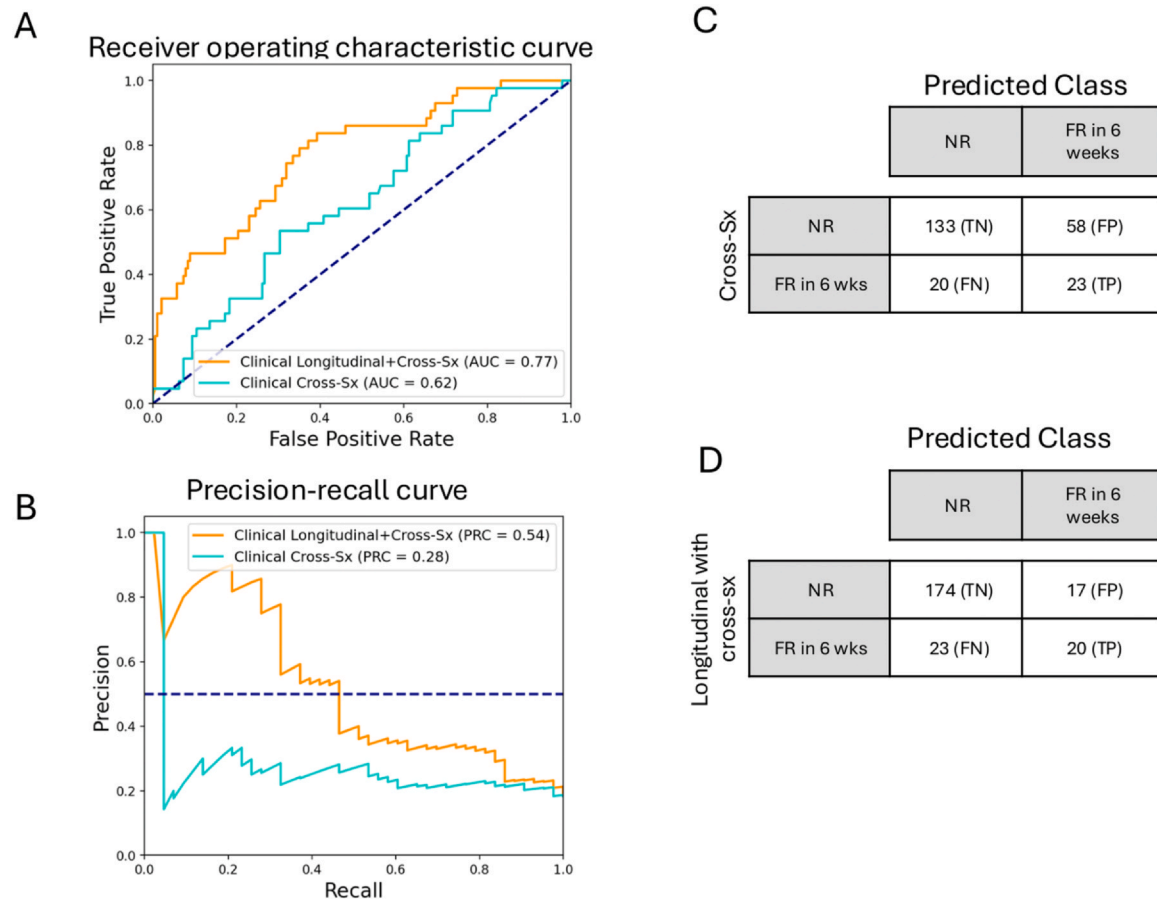


Figure 3 Performance of gradient-boosted models with stratified 10-fold cross-validation for cross-sectional-only vs longitudinal and cross-sectional combined clinical variables in predicting future CAR within the first 180 days post-HT. (A) AUROC and (B) AUPRC curves for cross-sectional-only (blue) and longitudinal/cross-sectional combined (orange) feature spaces. Confusion matrices for (C) cross-sectional-only predictors and (D) longitudinal/cross-sectional combined predictors are shown.

tissue metrics, with additional contributions from myocyte area ratio and immune cell indices. Among clinical variables, changes in immunosuppressive therapy (e.g., discontinuation

of mycophenolate mofetil) were among the most influential predictors, while laboratory and demographic variables contributed broadly but with lower magnitude.

Table 2 Performance Reporting of Models Predicting Rejection in Different Time Windows and With Different Methods

Time window	Test Set N	# Pos. (%)	Feature Space	AUROC	AUPRC	Macro F1	Sens.	Spec.	PPV	NPV	p-value
CLINICAL VARIABLES ONLY											
0-180d	234	43 (18.4)	Cross-Sx only	0.62	0.28	0.57	0.53	0.70	0.28	0.87	0.007 (ROC)
			Cross-Sx and longitudinal combined	0.77	0.54	0.70	0.47	0.91	0.54	0.88	0.0038 (PRC)
INTEGRATED CLINICAL-MORPHOLOGIC MODEL											
0-180d	234	43 (18.4)	Cross-Sx only	0.72	0.47	0.66	0.40	0.90	0.47	0.87	0.013 (ROC)
			Cross-Sx and longitudinal combined	0.86	0.74	0.83	0.72	0.94	0.74	0.94	0.011 (PRC)
CROSS-SECTIONAL AND LONGITUDINAL VARIABLES COMBINED											
0-180d	234	43 (18.4)	Clinical	0.77	0.54	0.70	0.47	0.91	0.54	0.88	0.057 (ROC)
			Integrated	0.86	0.74	0.83	0.72	0.94	0.74	0.94	0.029 (PRC)

*Abbreviations: AUPRC: area under the precision-recall curve; AUROC: area under the receiver operating characteristic curve; cross-sx: cross-sectional; NPV: negative predictive value; PPV: positive predictive value; Sens.: sensitivity; Spec.: specificity.

Bootstrapping with $N = 1,000$ samples was used for statistical comparison between curves.

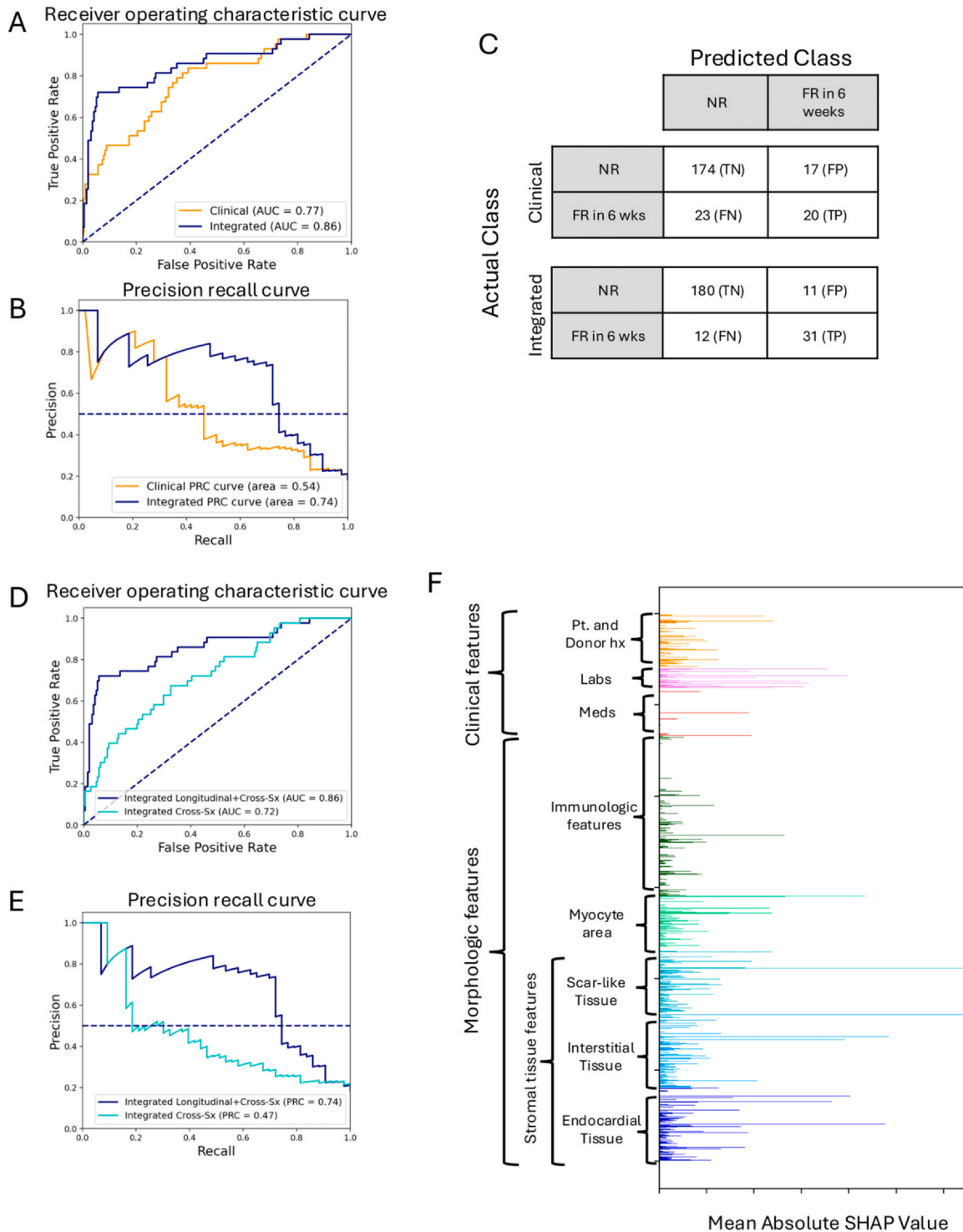


Figure 4 Performance and feature space of clinical-only and morphology-integrated models for predicting 6-week future CAR within the first 180 days post-HT. (A) receiver-operating curve, (B) precision-recall curve, and (C) confusion matrix for clinical-only (orange) vs morphology-integrated (dark blue) models. (D) receiver-operating curve and (E) precision-recall curve for cross-sectional-only integrated (light blue) and longitudinal/cross-sectional morphology integrated (dark blue) models. (F) SHAP (Shapley Additive Explanations) summary plot showing the relative contribution of features to the best-performing prediction model. Features are grouped into clinical domains (structured EHR data) and morphologic domains (quantitative features derived from digitized H&E EMB slides), with additional subcategorization by feature type. Bars represent mean absolute SHAP values across the dataset, with higher values indicating greater contribution to model predictions. Features are ranked by overall importance within the model.

Translating predictive signal into a clinically actionable risk index

To enable clinical translation of the high-dimensional machine-learning XGBoost model, we derived the IRRI, a simplified scoring system constructed from the most informative, interpretable, and routinely available predictors identified by SHAP analysis of the best-performing XGBoost model.

To preserve the added value of multi-modal integration while ensuring interpretability and ease-of-use, IRRI incorporates 2 complementary domains—Clinical and Morphologic—each comprising 4 criteria. Each domain yields a 0-4 composite score (Figure 5A), which are then combined to generate a final IRRI score. Clinical criteria capture renal dysfunction, liver function abnormalities, prior EMB grade, and recent immunologic risk factors, while morphologic criteria quantify endocardial thickening, interstitial expansion, stromal-to-myocyte ratio, and lymphocyte

infiltration. Details of morphologic feature derivation are provided in the [Supplemental Methods](#) and [Table S4](#). Clinical and morphologic composite score thresholds were defined using a pragmatic, non-optimized approach based on the number of risk criteria present (≤ 1 = low risk, 2 = intermediate risk, ≥ 3 = high risk), prioritizing interpretability and clinical usability. Predictive performance of these predefined groupings was subsequently evaluated in the held-out test set.

Predictive performance of IRRI risk domains

Logistic regression demonstrated that the presence of IRRI risk criteria was strongly associated with future CAR (Figure 5B). Patients with high clinical risk (composite score ≥ 3) had a log-odds ratio of 2.96 (95% CI: 2.00-3.91) relative to the low-risk group, corresponding to a 19.3-fold total increase in risk. High morphologic risk was also

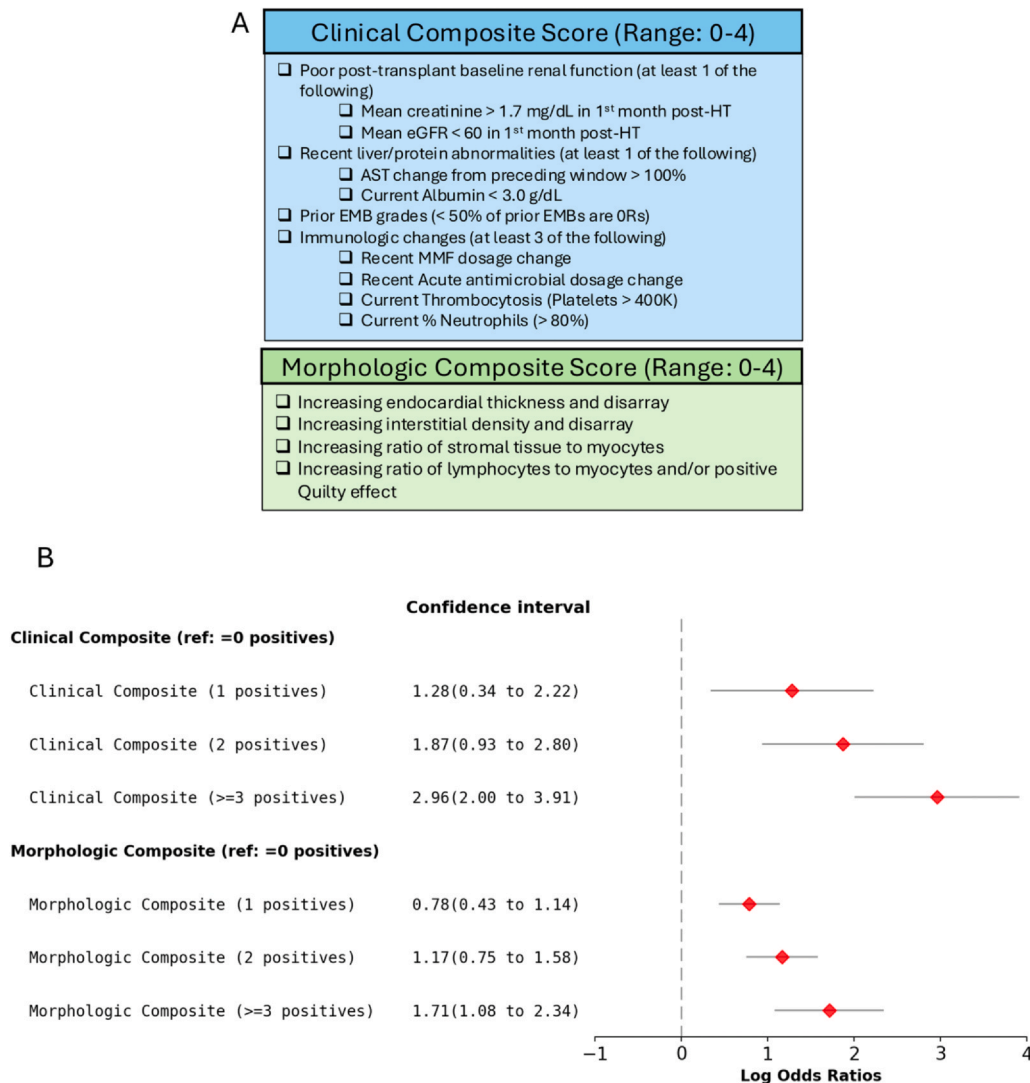


Figure 5 Development of an end-user-friendly approach to stratifying low- and high-risk clinical and morphologic patients. (A) Model-derived criteria for clinical and morphologic composite scores. (B) Log-odds scores of developing any future CAR within the first 180 days post-HT for different risk groups, compared to a reference of either 0 clinical composite positives or 0 morphologic composite positives, respectively.

independently predictive (log-odds ratio 1.71, 95% CI: 1.08–2.34), corresponding to a 5.5-fold increase in odds.

IRRI stratifies distinct future CAR risk trajectories

By combining clinical and morphologic domains, IRRI stratifies individual patient encounters into low-, intermediate-, and high-risk groups for CAR within 90 days of an outpatient visit (Figure 6E). Kaplan–Meier analysis demonstrated clear separation between strata (Figure 6A–C). High-risk encounters identified by integrated IRRI had a 75.6% CAR rate within 90 days, significantly exceeding rates observed using clinical-only (43.7%, $p < 0.001$) or morphologic-only (39.0%, $p < 0.001$) criteria. Conversely, the low-risk integrated IRRI group exhibited strong negative predictive value, with only 10.7% of encounters followed by CAR—significantly lower than clinical-only (17.7%, $p = 0.024$) or morphologic-only (26.1%, $p < 0.001$) low-risk classifications.

Cox proportional-hazards modeling confirmed these findings (Figure 6D). High-risk integrated IRRI encounters were associated with a hazard ratio of 6.15 (95% CI: 4.17–9.09) for future CAR, while intermediate-risk encounters showed a modest increase in risk (HR 1.65, 95% CI: 1.29–2.11). Low-risk encounters were protective (HR 0.52, 95% CI: 0.33–0.84). Additional descriptive analyses of demographic distributions and clinical outcomes across IRRI risk strata are provided in Supplemental Table S5. In exploratory analyses, IRRI risk categories remained independently associated with future CAR after adjustment for demographic and era-related factors (Supplemental Table S6) and demonstrated a preserved risk gradient when stratified by rejection phenotype, with the strongest effects observed for ACR and more limited inference for AMR due to low event counts (Supplemental Table S7).

Discussion

CAR remains a leading cause of morbidity and graft failure following HT. Here, we demonstrate the feasibility of predicting future CAR using machine learning applied to longitudinal clinical data and quantitative biopsy morphometrics from routinely acquired EMBs. By integrating laboratory trends, IS trajectories, and computationally extracted histology features, we developed a predictive framework capable of identifying high-risk patients weeks to months in advance of overt CAR. Importantly, we translated high-dimensional model outputs into IRRI—a simplified, interpretable, and high-performing risk stratification tool. Together, these findings establish a practical pathway toward personalized CAR surveillance and IS management in post-HT care.

Temporal dynamics and multi-domain integration drive predictive performance

A central insight from this work is that CAR risk is better captured by temporal trends than by static snapshots.

Rejection often emerges gradually, preceded by subtle, cumulative shifts in lab values, medication doses, and tissue morphology. Incorporating these dynamic features yielded substantial performance gains, ultimately reaching an AUROC of 0.86 with full multi-modal integration. These findings highlight the additive value of modeling patient trajectories over time.

Importantly, embedding biopsy morphometrics alongside clinical trends yielded superior predictive value compared to either domain alone. While previous studies examining the utility of clinical biomarkers for CAR prediction have been published,^{32,33} our findings underscore the importance of a multi-domain framework incorporating diverse and complementary predictors—including features not captured by standard CAR grading. Features such as endocardial stromal thickening, interstitial expansion, increased stromal-to-myocyte ratio, and subtle differences in overall lymphocyte counts were among the strongest independent predictors of future rejection. These results reinforce that EMBs remain a critical source of diagnostic and prognostic insight—especially when enhanced by computational methods that quantify histologic features reproducibly and at scale.

This study builds upon a growing body of artificial-intelligence (AI)-aided digital pathology research in HT. Prior studies have demonstrated that computational morphometrics could match expert pathology grading,²⁵ outperform conventional grading for assessing active rejection severity,²⁷ and be leveraged to provide predictive insights into long-term outcomes such as graft dysfunction and allograft vasculopathy.^{24,26,34} The present work represents the next phase in translational “AI” diagnostics in HT, emphasizing multi-domain data integration, Bayesian logic, and end-user-focused decision support.

Morphologic predictors reflect signals of subclinical immune injury

Notably, the top-performing morphologic predictors in our model likely reflect underlying biology associated with early immune activation and tissue remodeling, latent features which are not adequately captured in standard ISHLT grading. For example, increased stromal-to-myocyte ratio and interstitial expansion may represent extracellular matrix accumulation driven by low-grade inflammation. Endocardial thickening may reflect subendocardial matrix expansion, edema, or Quilty lesions associated with chronic alloimmune stimulation. Similarly, increased lymphocytic infiltration in the absence of meeting formal rejection criteria may signal ongoing but subclinical immune surveillance and adds to prior work which highlighted the long-term prognostic implications of recurrent “low-grade” rejection events.³⁴

These observations suggest that computational histology offers a window into the biologic continuum of rejection, capturing tissue-level signs of inflammation, damage, and repair before overt rejection is detected. This capability complements and enhances the utility of traditional EMB

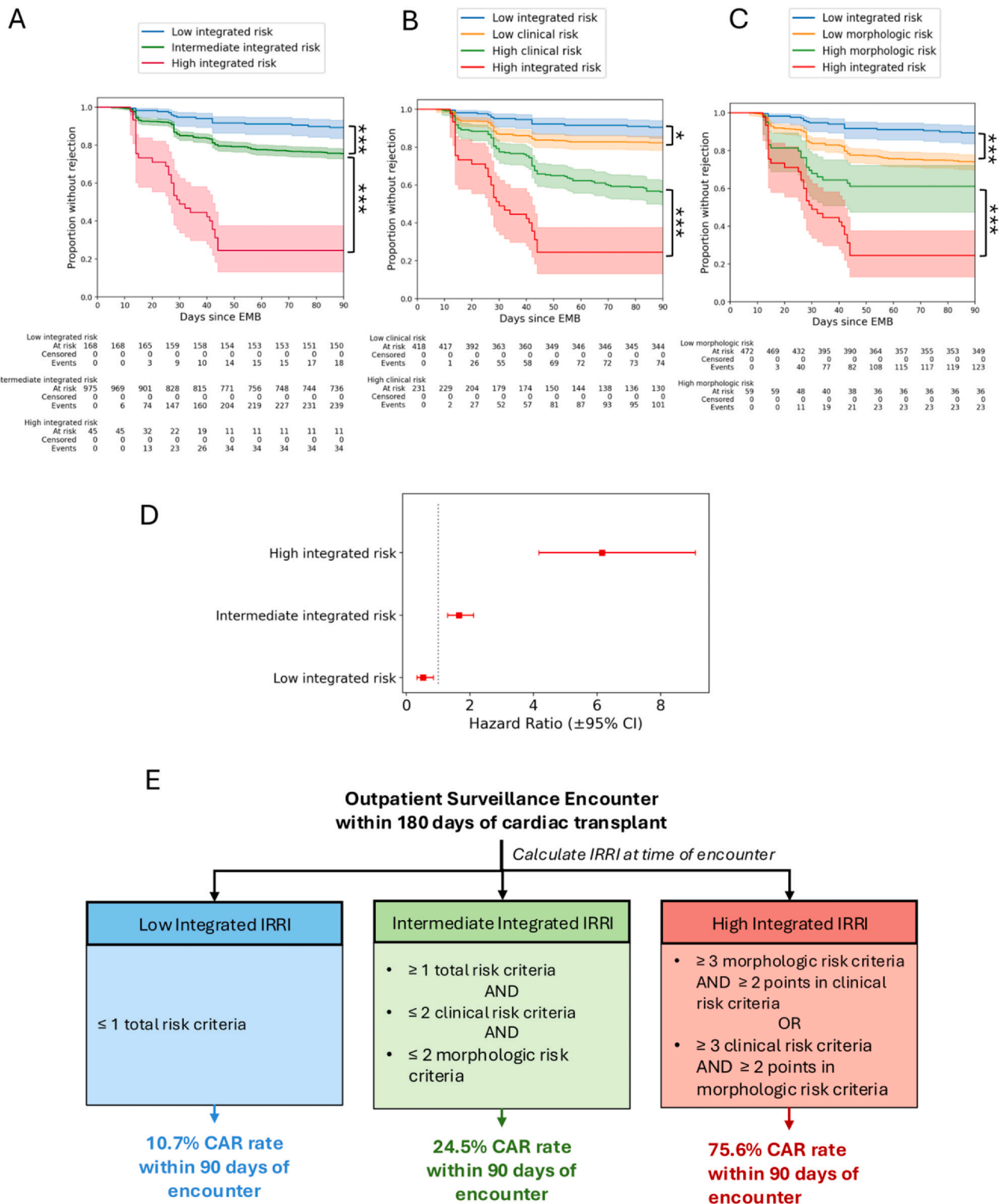


Figure 6 Time-to-rejection trajectory comparisons for clinical-only, morphologic-only, and integrated composite score risk groups. Encounter-level trajectory comparison between (A) low, intermediate, and high integrated risk groups, (B) low- and high-risk clinical-only and integrated risk groups, and (C) low- and high-risk morphologic-only and integrated risk groups. (D) Cox-regression hazard ratios of future CAR after an encounter between low, intermediate, and high integrated risk groups. (E) and (F) Patient-level stratification based on the highest-risk encounter during the study period. (E) Clinical decision workflow and IRRI criteria for risk stratification at time of outpatient surveillance encounter. Kaplan-Meier curves compared for statistical significance using log-rank testing and proportions compared using Z-test for proportions ($*p < 0.05$, $**p < 0.01$, $***p < 0.001$).

grading, offering the potential to shift transplant care from reactive to proactive.

From model performance to practical translation: The role of IRRI

Past studies utilizing ML approaches to transplant rejection generally focus on model performance and optimization, often at the expense of interpretability and clinical usability.^{35–39} In contrast, the present study explicitly sought to bridge the gap between high-dimensional predictive modeling and real-world clinical decision-making. To this end, we combined data-driven feature selection with clinical domain expertise to derive the IRRI, a simplified and interpretable risk stratification tool constructed from 8 key predictors—4 clinical and 4 morphologic—that are routinely available in post-transplant care.

Despite its simplicity relative to the full XGBoost model, IRRI preserved strong predictive performance. Encounters classified as high risk demonstrated a positive predictive value of 75.6% for CAR within 90 days, while low-risk encounters exhibited a negative predictive value of 89.3%. Although our cohort was enriched for CAR events (~1.8-fold higher incidence vs background FR rate), which likely influences absolute IRRI risk estimates as well as positive and negative predictive values, the magnitude of relative risk separation between high- and low-risk groups is statistically significant and clinically meaningful. These findings suggest that IRRI captures clinically meaningful gradients of rejection risk that will remain significant regardless of cohort composition.

From a translational perspective, IRRI provides a framework for risk-adaptive CAR surveillance and IS strategies. Although more work needs to be completed to determine IRRI's generalizability to external institutions and larger populations, the potential for utilizing a data-driven decision-support tool such as IRRI in stratifying CAR risk warrants further discussion. Low-risk patients may safely defer some routine testing, reducing procedural burden, patient anxiety, and healthcare costs. Conversely, high-risk patients may benefit from closer monitoring, delayed IS tapering, or preemptive therapeutic adjustment prior to overt CAR onset. Importantly, IRRI may also inform the use of emerging non-invasive biomarkers such as donor-derived cfDNA. In patients identified as high risk by IRRI - where pre-test probability of rejection is already elevated - deferring cfDNA testing may be appropriate, as screening assays are less informative in this setting and may delay definitive EMB evaluation while increasing cost.

More broadly, IRRI offers a high-specificity contextual framework for interpreting cfDNA results, which are sensitive but prone to non-specific elevations. By integrating tissue-derived morphologic signals with clinical trajectories, IRRI may help distinguish biologically meaningful graft injury from false-positive cfDNA screening results. This approach parallels long-standing practice in coronary artery disease, where pre-test probability modeling guides the choice between no further testing, non-invasive evaluation,

or definitive invasive angiography. Analogously, IRRI may help determine which HT patients are candidates for surveillance reduction vs protocolized cfDNA screening vs direct biopsy-based evaluation, supporting more efficient and more personalized post-transplant care.

Limitations and future directions

Several limitations warrant acknowledgment. First, while a large cohort, this was a single-center study, and findings may reflect local practice patterns and protocols as well as local populations. While the quantitative morphology pipeline was externally validated previously,²⁵ external validation of IRRI across diverse institutions will be necessary to establish generalizability. Second, real-world implementation of IRRI will require integration with clinical workflows, including automated data extraction from the EHR and biopsy digitization pipelines. Third, temporal information was represented using structured time bins rather than higher-resolution sequential modeling approaches. While this strategy preserves interpretability and aligns with our goal of deriving a simplified clinical risk framework, it may obscure shorter-term fluctuations relevant to acute rejection events. More complex approaches, such as recurrent neural networks, could potentially improve temporal fidelity but would introduce tradeoffs in interpretability, computational burden, and ease of clinical implementation. Fourth, while we considered the HT cohort as an aggregate for modeling, future work should be pursued to de-aggregate the cohort and assess socioeconomic variables in post-HT CAR risk and adverse outcomes. Finally, the number of EMBs per patient was limited by availability of digitized slides rather than total biopsy burden, which may reduce temporal resolution. However, given current practice trends towards less frequent EMB sampling, our study cohort may better reflect the structure of contemporary transplant practice better than a cohort with 5-6 early post-transplant EMBs would have.

Future research efforts should focus on several complementary tasks. First, incorporation of additional data modalities, including donor-derived cell-free DNA and unstructured echocardiographic and electrocardiographic data, may further refine IRRI risk assessments and enable more comprehensive multi-modal prediction frameworks. Second, achieving meaningful clinical impact will require prospective cohorts and implementation studies to determine whether IRRI-guided personalization of CAR surveillance can safely reduce testing burden, adverse events, and/or healthcare costs. Real-world implementation will additionally require integration with digital pathology and EHR infrastructures, with attention to workflow integration, data standardization, and regulatory considerations.

Conclusion

This study demonstrates that machine learning applied to routine post-transplant data can generate high-performing,

interpretable models for future rejection prediction. By integrating longitudinal clinical features with quantitative biopsy morphometrics and distilling these into the IRR scoring system, we offer a practical framework for personalized CAR surveillance and IS management.

Disclosures

None to state.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.healun.2026.04.031](https://doi.org/10.1016/j.healun.2026.04.031).

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